

Cheap to Verify, Expensive to Fix: Deterministic Structural Verification as a Routing Signal for Matched-Cost VLM Table Extraction

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Abstract—Vision-language models (VLMs) used for table extraction spend the same visual tile budget on every page, regardless of difficulty. We study a simple alternative: parse each page once at a low tile budget, check the emitted HTML with a deterministic structural verifier (parseability, span consistency, grid rectangularity, table presence), and reparse the page once at a higher budget only if the check fails. The verifier requires no training, no OCR, and no reference output, and each routing decision can be audited by re-running a parser. We evaluate this policy under a matched-cost protocol in which the adaptive system, a fixed-budget baseline, and a random-escalation control matched in expected cost all spend a near-equal mean tile budget per page, with budgets frozen on separate calibration splits. On held-out FinTabNet pages ($n=150$, InternVL2-2B), verifier-gated escalation reaches 0.133 macro cell-F1 versus 0.120 for the cost-matched fixed baseline, and its advantage over random escalation at equal expected cost is the only contrast that remains significant after Holm correction (paired Wilcoxon, adjusted $p=0.0007$). The gain persists against a post-hoc 11-tile baseline given more budget than the adaptive system, although the fixed-baseline contrast itself stays borderline ($p \approx 0.05-0.06$). On held-out OmniDocBench pages ($n=90$) the policy shows no measurable gain: escalation becomes nearly uniform (78% of pages) and adaptive, random, and fixed systems are statistically indistinguishable. The contrast suggests that structural verification is a useful routing signal precisely where uniform budget increases have weak returns, and adds little where most pages need the higher budget anyway. We report verifier failure-code distributions, per-page reparse outcomes, and wall-clock measurements showing that matched tile budgets do not imply matched latency.

Index Terms—vision-language models, adaptive inference, table extraction, structural verification, matched-cost evaluation, document understanding

I. INTRODUCTION

A VLM-based table extractor commits the same per-page compute whether the page holds a short list as opposed to a dense financial statement with merged cells. Modern open VLMs such as InternVL2 expose this compute as a visual tile budget: the page image is split into up to B fixed-resolution tiles before encoding, so the budget is an explicit, per-request knob [1], [2]. Fixed-budget inference spends most of that budget on pages that did not need it.

Table extraction has a property that makes a cheap routing signal plausible: the output format has crisp syntactic invariants. A predicted HTML table either parses or it does not; its `rowspan`/`colspan` declarations either expand into a rectangular cell grid or they do not; a page either contains a `<table>` element or it does not. Checking these invariants takes under two milliseconds per page in our measurements, while a single 2B model pass takes tens of seconds on the

same hardware: verification is cheap; producing a better parse is expensive.

This paper addresses a specific question: can a deterministic structural check on the model’s own output route a second, higher-budget parse to the pages that need it, well enough to beat spending the same average tile budget uniformly, and more stringently, to beat spending it randomly? The random-escalation control matters as much as the fixed baseline. Any escalation policy buys extra compute for some pages; if random escalation at the same probability does as well as the verifier, the verifier signal carries no information. Our protocol therefore compares the adaptive system against fixed-budget baselines *and* a random control matched in expected tile cost, with all budgets frozen on calibration data before the held-out evaluation.

We make three contributions.

- 1) **An auditable, training-free routing mechanism.** A deterministic verifier over emitted table HTML (five failure codes; Section III-B) gates a single full-page reparse at a higher tile budget. There is no learned router, no OCR on the trigger path, and at most one reparse per page; every routing decision can be reproduced from the page’s raw output alone.
- 2) **A matched-cost evaluation protocol with a random control.** Budgets are calibrated and frozen on separate 50-page splits per dataset; the held-out comparison includes a low-budget floor, a cost-matched fixed baseline, a random-escalation control at the calibration-measured escalation rate (three seeds), a tile-matched 8B reference, and a post-hoc over-budget fixed baseline that brackets the adaptive system’s realized cost from above. Paired tests are reported with Holm correction across the confirmatory family.
- 3) **A conditional empirical result on two datasets.** On FinTabNet (150 held-out pages) the verifier-gated policy beats random escalation at equal expected cost (the only Holm-significant contrast) and shows a consistent $+0.014$ cell-F1 margin over fixed baselines at both 10 and 11 tiles. On OmniDocBench (90 held-out pages) the policy shows no gain over either control. The budget-response curves of the two datasets explain the split: uniform budget increases barely help FinTabNet but help OmniDocBench a great deal, leaving selective escalation no room to win there.

We also report what fixed-budget evaluations usually omit: verifier failure-code distributions, per-page reparse outcomes (improved, unchanged, regressed), per-seed random-control variance, and wall-clock latency, which shows that matching tile budgets does not match latency.

II. RELATED WORK

VLM document and table parsing. End-to-end vision-language models have largely replaced detect-then-recognize cascades for document parsing. Donut [3] and Pix2Struct [4] treat parsing as conditional generation from pixels; open VLM families such as InternVL [1] provide size-paired checkpoints (2B/8B) [5] and a dynamic tiling mechanism [2] that makes visual compute an explicit budget. Table-structure work, including TableFormer [6] and PubTables-1M [7], together with the PubTabNet benchmark and its TEDS metric [8], TableBank [9], and FinTabNet [10], established the datasets and metrics we build on. Throughout this line of work, inference is fixed-budget: no mechanism decides per page where extra visual budget should go.

Adaptive computation. Spending more compute on harder inputs has a long history, from cascaded detectors [11] and early-exit networks [12] to adaptive computation time [13], confidence-based early exit in language models [14], speculative decoding [15], and cost-aware model cascades [16]. These mechanisms typically rely on a *learned* signal, such as a halting unit, a confidence head, or a trained router. Our escalation signal is instead a deterministic predicate on the structured output itself, which removes training-data requirements and makes individual routing decisions inspectable.

Output verification and self-correction. Verifier-guided generation usually employs a learned critic: trained solution verifiers for math [17], process supervision [18], model self-critique [19], or agreement-based selection over samples [20]. A learned critic can judge content but is itself a model with failure modes and training costs. We restrict the checker to structural well-formedness; it can only declare an output structurally acceptable or not, and it cannot rewrite anything. This narrow contract is what makes the check deterministic, near-zero-cost, and auditable.

The gap this paper addresses sits at the intersection: fixed-budget VLM table extraction does not allocate visual budget per page; adaptive computation methods allocate but need learned signals; verification work supplies signals but usually learned ones. We evaluate the simplest such signal, a deterministic structural predicate used as the routing signal, under matched-cost controls designed to detect whether the signal itself, rather than the extra compute, produces any gain.

III. VERIFIER-GATED ADAPTIVE INFERENCE

Figure 1 shows the pipeline. We describe the backbone and prompt, the verifier, the escalation policy, and the cost model that the matched-cost protocol relies on.

A. Setting and backbone

We use InternVL2-2B for all 2B systems and InternVL2-8B for the 8B reference, loaded from the public checkpoints [5] with bfloat16 weights on a single NVIDIA L4 (24 GB). InternVL2 splits each page image into at most B fixed-resolution tiles before visual encoding, so B (“max tiles”) upper-bounds visual compute per pass; under a fixed prompt and greedy decoding it is the only per-request compute knob we vary. All systems share one fixed prompt that instructs the model to transcribe the page and emit every table as raw HTML `<table>` markup, explicitly forbidding markdown pipe tables and bulleted reformatting; decoding is greedy with a 2,048-token cap, no sampling anywhere. Because decoding is greedy, the adaptive system’s first pass is bit-for-bit the low-budget baseline: across all 240 held-out pages the two produced identical output token counts.

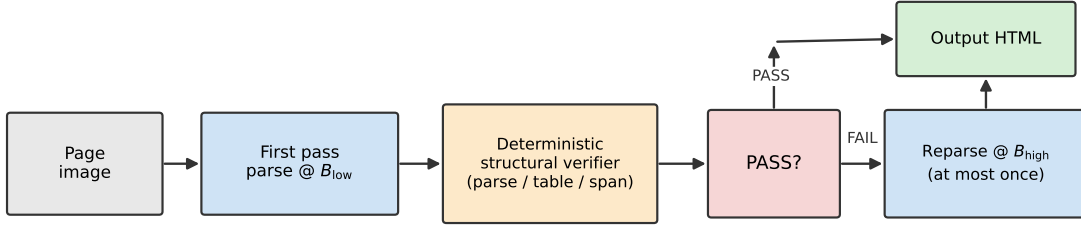
One implementation detail affects runtime and is disclosed for reproducibility. On hard pages the 2B model sometimes degenerates into an exactly periodic token loop and runs to the 2,048-token cap. The adapter detects exact periodicity over a trailing 100-token window (period ≤ 50) and truncates the repeating tail. The detector never alters which tokens are chosen; it only stops obvious loops early. Re-running the frozen calibration points under it left tile budgets unchanged (one borderline FinTabNet page changed its reparse decision, shifting measured adaptive cost from 10.16 to 10.48 tiles per page; the frozen artifact records the re-measured value, and the random control’s escalation probability is derived from it).

B. Deterministic structural verifier

The verifier is a pure function from the raw output string to a decision in {PASS, REPARSE} plus a set of failure codes. It extracts `<table>` blocks from the output, then for each block attempts HTML parsing, expands `rowspan/colspan` declarations into a flat cell grid, and checks the grid. Five failure codes exist:

- `NO_TABLE_FOUND`, the output contains no `<table>` block;
- `HTML_PARSE_ERROR`, a block exists but does not parse;
- `SPAN_EXPANSION_FAILED`, span attributes cannot be resolved into a grid;
- `RECTANGULAR_INCONSISTENCY`, after span expansion, row lengths disagree;
- `DEGENERATE_TABLE`, the grid is empty or all cells are blank.

Any failure code on any block yields REPARSE. The verifier consults no reference output, no model, and no OCR; it performs no semantic checks, so a well-formed table with wrong content passes. It is intentionally a high-precision trigger for *structural* breakage, not a correctness oracle. Measured on the held-out runs, verification costs at most 1.6 ms per page (median 0.04–0.39 ms), against a mean of 17–33 s per model pass on the same machine: negligible relative to inference, though not literally free.



One forward pass on every page; a second pass only on verifier-flagged pages.

Fig. 1. The adaptive pipeline. Every page is parsed once at the low tile budget B_{low} . A deterministic structural verifier inspects the emitted HTML and either accepts the first pass or triggers exactly one reparse at B_{high} , whose output then replaces the first pass. There is no learned routing, no OCR, and no second reparse.

C. Escalation policy

The policy is one rule: if the verifier returns REPARSE, run the same model on the same full page once more at B_{high} , and keep the second output unconditionally; otherwise keep the first. At most one reparse per page is permitted, enforced structurally by the runner. We deliberately add no heuristics on top: no output-length checks, no retry on the second pass, and no selection between the two outputs. Keeping the replacement unconditional means a reparse *can* in principle produce a worse parse than the first pass; whether that happens is an empirical question we report on in Section V-D.

D. Cost model

Our pre-registered cost unit is the per-page tile budget. A fixed system at budget B costs B tiles per page. The adaptive system costs

$$c_i = B_{\text{low}} + r_i B_{\text{high}}, \quad r_i \in \{0, 1\}, \quad (1)$$

where r_i indicates a reparse on page i ; the mean cost over a split is $B_{\text{low}} + \bar{p} B_{\text{high}}$ with $\bar{p}$ the empirical reparse rate. Tile budget is the right *control* variable; it is the knob the systems actually differ on, it is deterministic, and visual-encoder compute scales with it; but it is not wall-clock latency and it is not FLOPs. Two passes carry fixed overheads, decode length varies, and an 8B model at the same tile count performs roughly four times the work per token of a 2B model. We therefore match systems on tiles, state so explicitly, and report measured wall-clock alongside (Section V-F); we avoid the phrase “equal compute.”

IV. EXPERIMENTAL DESIGN

A. Datasets, filtering, and splits

We evaluate on English table pages from two sources with deliberately different character.

OmniDocBench [21] provides diverse-source PDF pages with gold HTML table annotations. We keep pages annotated as English with at least one table whose gold cells include at least four non-empty entries. The filter exists because

the source occasionally tags layout scaffolding (slide bullet groups, mostly-empty forms) as tables; such pages score near zero for every system regardless of quality and only add noise. The filtered English-table population is 146 pages; we use 140: 50 for calibration and 90 held out.

FinTabNet, introduced as part of the GTE framework [10], contains financial-report pages with dense, span-heavy tables. We draw from the HTML-paired distribution of its test split, cap at 250 candidate pages, apply the same four-cell filter, and stratify-sample 200 pages by a difficulty bucketing (below): 50 for calibration, 150 held out.

Difficulty buckets are computed from gold metadata: *simple* (fewer than 6 rows), *complex* (6–15 rows), *very complex* (16+ rows or any merged cell). The held-out splits skew hard: 51 of 90 OmniDocBench pages and 91 of 150 FinTabNet pages are very complex. Calibration and held-out splits are disjoint; budgets were never re-selected on held-out pages.

B. Budget calibration

For each dataset independently, a sweep on the 50-page calibration split selects the budget pair $(B_{\text{low}}, B_{\text{high}})$ whose measured mean adaptive cost (Eq. 1) lies closest to a target of 10 tiles per page, then freezes the matched fixed budgets at the nearest integer to the measured adaptive cost. The frozen values are shown in Table I. Because fixed budgets live on an integer grid, exact equality with the adaptive system’s realized cost is not attainable; on FinTabNet we therefore additionally evaluate a post-hoc 11-tile fixed baseline that brackets the adaptive system’s held-out cost (10.37 tiles) from above (Section V-C).

C. Systems

Seven runs per dataset, identical prompt, weights per model size, decoding, page order, and scorer:

- **fixed_2b_low**: single pass at B_{low} . The floor, and (by greedy determinism) identical to the adaptive system’s first pass.

TABLE I
FROZEN PER-DATASET BUDGETS (MAX TILES) AND
CALIBRATION-MEASURED QUANTITIES. THE RANDOM CONTROL’S
PER-PAGE ESCALATION PROBABILITY $\hat{p}$ IS DERIVED FROM THE FROZEN
ARTIFACT AS $(\bar{c}_{\text{cal}} - B_{\text{low}})/B_{\text{high}}$.

	OmniDoc.	FinTab.
$B_{\text{low}} / B_{\text{high}}$	2 / 12	6 / 16
B^{fix} (2B and 8B)	11	10
Measured adaptive cost $\bar{c}_{\text{cal}}$	10.64	10.48
Reparse rate	0.72	0.28
Random control $\hat{p}$	0.72	0.28

- `fixed_2b_matched`: single pass at B^{fix} , the uniform-spending competitor at matched mean tile cost.
- `adaptive_2b`: the pipeline of Section III.
- `random_2b_seed{0, 1, 2}`: identical two-pass shape, but the verifier decision is discarded and replaced by an independent Bernoulli($\hat{p}$) draw with $\hat{p}$ from Table I. This control separates the value of the *signal* from the value of the *extra compute*: if the verifier carries no information, adaptive and random should perform alike at equal expected cost.
- `fixed_8b_matched`: InternVL2-8B, single pass at B^{fix} . A reference for “spend the budget on a bigger model”, matched in tiles, not in FLOPs (Section III-D).

D. Metrics

Macro cell-F1 (primary). Predicted and gold HTML are parsed with the same span expander the verifier uses; each cell becomes a tuple (table index, row, column, whitespace-normalized text), and per-page F1 is computed over exact tuple matches, then averaged unweighted across pages. Pages whose predicted HTML fails parsing or grid expansion score zero; malformed output is penalized, not repaired. The metric is strict: a table that is shifted by one column scores near zero even if its text is right.

TEDS (secondary) [8]: tree-edit-distance similarity over the parsed table trees, following the reference implementation’s structure. TEDS gives partial credit where cell-F1 does not, so reporting both bounds the metric sensitivity of our conclusions.

Parse-error count: pages whose prediction failed HTML parsing or span expansion outright.

All means carry 95% bootstrap confidence intervals (percentile, 1000 resamples).

E. Statistical protocol

We pre-specify four confirmatory paired comparisons on per-page cell-F1, namely {adaptive vs. fixed-matched, adaptive vs. random} $\times$ {OmniDocBench, FinTabNet}, tested with two-sided Wilcoxon signed-rank tests and Holm correction across the family of four. For the random control, the three seeds are pooled by averaging each page’s cell-F1 across seeds before pairing, so each page contributes one pair (no pseudo-replication); per-seed means and their spread are reported separately. Pairs with zero difference are dropped, per the standard zero-method; because many hard pages score zero

under every system, the number of informative (non-tied) pairs is small, and we report it for every test. The post-hoc 11-tile comparison is labeled exploratory and is not part of the corrected family; for it we also report a bootstrap CI on the mean paired difference.

V. RESULTS

A. Main matched-cost comparison

Table II reports all systems on both held-out splits; Figure 2 plots the same data against tile budget. Realized mean tile costs confirm the protocol: on OmniDocBench the adaptive system spent 11.33 tiles per page against the fixed baseline’s 11.0, and on FinTabNet 10.37 against 10.0. The adaptive system was, if anything, slightly over-budgeted, which makes its OmniDocBench null result conservative and motivates the over-budget check of Section V-C for FinTabNet.

Three observations are relevant to the results that follow. First, on FinTabNet, uniform budget increases do not help: the 6-tile floor (0.130) already exceeds the 10-tile (0.120) and 11-tile (0.120) fixed systems. Second, on OmniDocBench the opposite holds; cell-F1 triples from 2 to 11 tiles (0.039 to 0.120), and only the 8B model (0.200) rises above the 2B cluster. Third, the adaptive system is the best 2B system on FinTabNet (0.133) while reducing parse errors on both datasets (45 vs. 57 on OmniDocBench, 30 vs. 37 on FinTabNet relative to fixed-matched).

B. Does the verifier signal beat its controls?

Table III reports the confirmatory paired comparisons. One contrast survives Holm correction: on FinTabNet, the adaptive system beats the pooled random-escalation control by +0.0138 cell-F1 at equal expected tile cost (adjusted $p = 7.0 \times 10^{-4}$). Random escalation at the same rate does not reproduce the adaptive gain; per-seed means are 0.116, 0.119, 0.124 (s.d. 0.004) against the adaptive 0.133, so on this dataset the verifier’s choice of *which* pages to escalate carries measurable information beyond the escalation budget itself.

The FinTabNet adaptive-versus-fixed-matched contrast does not survive correction ($p = 0.048$ raw, 0.144 adjusted) and we do not claim it as significant. Its effect size, however, is consistent: +0.0137 against the 10-tile baseline and +0.0139 against the post-hoc 11-tile baseline, with a bootstrap CI on the latter difference excluding zero. With 137 of 150 pairs tied (mostly pages scoring zero under both systems), the rank test is powered by 13 informative pages; the borderline p-values reflect that, and a larger evaluation would be needed to settle the fixed-baseline contrast either way.

On OmniDocBench both confirmatory contrasts are null ($p \geq 0.27$ raw). Adaptive, random, and fixed-matched are statistically indistinguishable there, and we treat this as a finding rather than a failure: the policy escalated 70 of 90 pages (78%), so it nearly reproduces uniform high-budget spending, and no escalation rule that fires on almost every page can beat either uniform or random spending at the same rate.

TABLE II

HELD-OUT RESULTS. COST IS MEAN VISUAL TILES PER PAGE (REALIZED, FROM EQ. 1 FOR TWO-PASS SYSTEMS). CELL-F1 AND TEDS ARE MACRO MEANS WITH 95% BOOTSTRAP CIs; PARSE-ERR COUNTS PAGES WHOSE PREDICTION FAILED PARSING OR GRID EXPANSION, WITH ITS PERCENTAGE OF THE n HELD-OUT PAGES IN PARENTHESES. THE RANDOM ROWS SHOW EACH SEED, SINCE THE THREE SEEDS ARE SEPARATE RUNS. THE 11-TILE FINTABNET BASELINE IS THE POST-HOC OVER-BUDGET CHECK OF SECTION V-C.

Dataset	System	Cost	Macro cell-F1 [95% CI]	TEDS [95% CI]	Parse-err (%)
OmniDocBench (n=90)	fixed 2B low (2 tiles)	2.00	0.039 [0.010, 0.075]	0.106 [0.067, 0.152]	70 (78%)
	fixed 2B matched (11 tiles)	11.00	0.120 [0.070, 0.173]	0.268 [0.214, 0.325]	57 (63%)
	adaptive 2B	11.33	0.121 [0.071, 0.175]	0.274 [0.214, 0.333]	45 (50%)
	random 2B seed 0	11.20	0.123 [0.074, 0.179]	0.249 [0.195, 0.310]	55 (61%)
	random 2B seed 1	11.47	0.120 [0.069, 0.172]	0.245 [0.186, 0.300]	51 (57%)
	random 2B seed 2	10.13	0.100 [0.055, 0.149]	0.221 [0.163, 0.280]	61 (68%)
	fixed 8B matched (11 tiles)	11.00	0.200 [0.132, 0.271]	0.396 [0.326, 0.469]	38 (42%)
FinTabNet (n=150)	fixed 2B low (6 tiles)	6.00	0.130 [0.089, 0.174]	0.483 [0.435, 0.527]	41 (27%)
	fixed 2B matched (10 tiles)	10.00	0.120 [0.078, 0.164]	0.465 [0.416, 0.508]	37 (25%)
	fixed 2B over-budget (11 tiles)	11.00	0.120 [0.078, 0.164]	0.462 [0.414, 0.506]	37 (25%)
	adaptive 2B	10.37	0.133 [0.092, 0.179]	0.492 [0.447, 0.535]	30 (20%)
	random 2B seed 0	9.95	0.116 [0.075, 0.159]	0.462 [0.417, 0.506]	41 (27%)
	random 2B seed 1	9.84	0.119 [0.080, 0.162]	0.476 [0.430, 0.519]	44 (29%)
	random 2B seed 2	11.55	0.124 [0.085, 0.169]	0.472 [0.426, 0.517]	42 (28%)
	fixed 8B matched (10 tiles)	10.00	0.134 [0.092, 0.182]	0.527 [0.481, 0.573]	46 (31%)

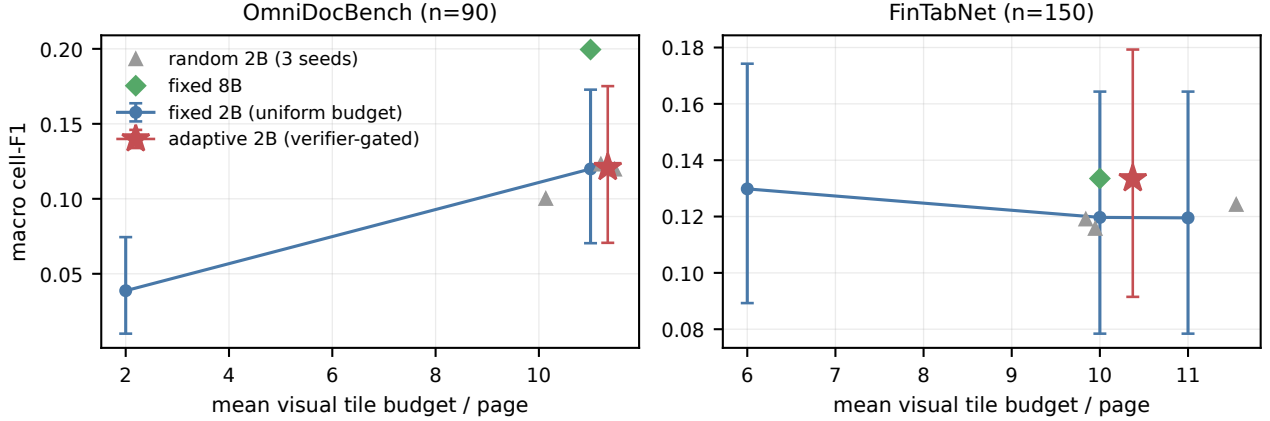


Fig. 2. Macro cell-F1 against mean tile budget. The two datasets respond to uniform budget in opposite ways. OmniDocBench (left) rises steeply from 2 to 11 tiles, and the adaptive policy lands on the fixed-budget line: selective escalation has nothing to add when uniform spending already pays. FinTabNet (right) is flat to slightly negative from 6 to 11 tiles; more uniform budget does not help. Yet the verifier-gated system sits above every same-cost 2B alternative and at the level of the tile-matched 8B reference. Error bars are *marginal* per-system 95% bootstrap CIs (percentile, 1000 resamples); the gray triangles plot all three random-control seeds individually, each the per-seed mean over pages, not a pooled mean. The significant adaptive-versus-random result is a *paired* per-page test (Table III) and is not represented by the overlap of these marginal intervals.

C. The over-budget check

Because the adaptive system’s realized FinTabNet cost (10.37 tiles) exceeds the frozen fixed baseline (10.0), its margin could be attributed to the extra 0.37 tiles rather than to routing. After the held-out results were known, we ran one additional fixed baseline at 11 tiles, *more* budget than the adaptive system used, under the identical prompt, weights, decoding, and scorer, and we report it regardless of outcome. The extra tile bought nothing: 0.1195 cell-F1 versus 0.1197 at 10 tiles, with parse errors unchanged (37). The adaptive margin persists (+0.0139, CI [+0.002, +0.029]), bracketing the adaptive cost from both sides: spending *less* than adaptive (10 tiles), the same uniform policy scores 0.120; spending

more (11 tiles), 0.120; routing the same average budget with the verifier, 0.133. We label this comparison exploratory; it was chosen post hoc, and its rank-test p-value (0.058) is as borderline as the 10-tile one. But it removes the budget-mismatch explanation for the FinTabNet effect: on this dataset the gain tracks *where* tiles were spent, not how many.

D. Verifier diagnostics

Table IV and Figure 3 open the routing decisions to inspection. Two of the five failure codes fired on held-out data. The trigger profile differs by dataset in an interpretable way: OmniDocBench failures are almost entirely missing tables (69 of 70); at 2 tiles the model often transcribes the page without emitting `<table>` markup at all. FinTabNet, whose first pass

TABLE III

PAIRED WILCOXON TESTS ON PER-PAGE CELL-F1. THE FIRST FOUR ROWS ARE THE PRE-SPECIFIED CONFIRMATORY FAMILY (HOLM-CORRECTED); THE LAST ROW IS THE POST-HOC OVER-BUDGET CHECK (EXPLORATORY, UNCORRECTED, WITH A BOOTSTRAP CI ON THE MEAN PAIRED DIFFERENCE). “INF. PAIRS” COUNTS NON-TIED PAIRS ACTUALLY ENTERING EACH TEST; TIES AT ZERO ARE COMMON BECAUSE MANY VERY-COMPLEX PAGES SCORE 0 UNDER EVERY SYSTEM.

Dataset	Comparison	Δ	Inf. pairs	p	p_{Holm}
FinTab	adapt. vs. random	+0.0138	29/150	1.8×10^{-4}	7.0×10^{-4}
FinTab	adapt. vs. fixed 10	+0.0137	13/150	0.048	0.144
OmniDoc	adapt. vs. random	+0.0066	26/90	0.269	0.538
OmniDoc	adapt. vs. fixed 11	+0.0009	32/90	0.729	0.729
FinTab	adapt. vs. fixed 11	+0.0139	51/150	0.058	—
bootstrap 95% CI of mean difference: [+0.002, +0.029]					

TABLE IV

VERIFIER BEHAVIOR ON HELD-OUT FIRST PASSES, AND THE OUTCOME OF EVERY TRIGGERED REPARSE ON PER-PAGE CELL-F1. OUTCOMES ARE OBSERVATIONS, NOT GUARANTEES: THE REPARSE OUTPUT REPLACES THE FIRST PASS UNCONDITIONALLY.

	OmniDoc.	FinTab.
Pages (held-out)	90	150
Reparse triggered	70 (78%)	41 (27%)
NO_TABLE_FOUND	69	28
RECTANGULAR_INCONSISTENCY	1	13
other three codes	0	0
Reparse improved cell-F1	20	6
Reparse left cell-F1 unchanged	50	35
Reparse lowered cell-F1	0	0

already runs at 6 tiles, instead mixes missing tables (28) with span-expansion inconsistencies (13), consistent with partially malformed complex tables.

The reparse yield is modest. On OmniDocBench, 20 of 70 reparses improved the page’s cell-F1; on FinTabNet, 6 of 41. The remaining triggered pages were structurally broken at both budgets, so the reparse spent tiles without measured benefit. These are correctly flagged pages the higher budget could not fix, an escalation-ceiling property of the backbone rather than a verifier error in the usual sense. No reparse lowered cell-F1 in any of the 111 triggered pages across both datasets. We emphasize that this is an empirical observation: the pipeline keeps the second output unconditionally, so regression is possible by design and simply did not occur in these runs. False negatives (pages that passed but would have gained from a reparse) cannot be counted without reparsing every page, which the protocol forbids; the OmniDocBench fixed-matched results imply such pages exist there, since uniform escalation lifted many pages the verifier passed.

E. Difficulty stratification

Table V localizes both the headroom and the ceiling. On FinTabNet, raising the uniform budget from 6 to 10 tiles *hurts* simple pages (0.190 to 0.142) and does nothing for the rest, while the adaptive system preserves the floor’s scores where the floor was already right and gains slightly on very-complex pages. On OmniDocBench the uniform increase

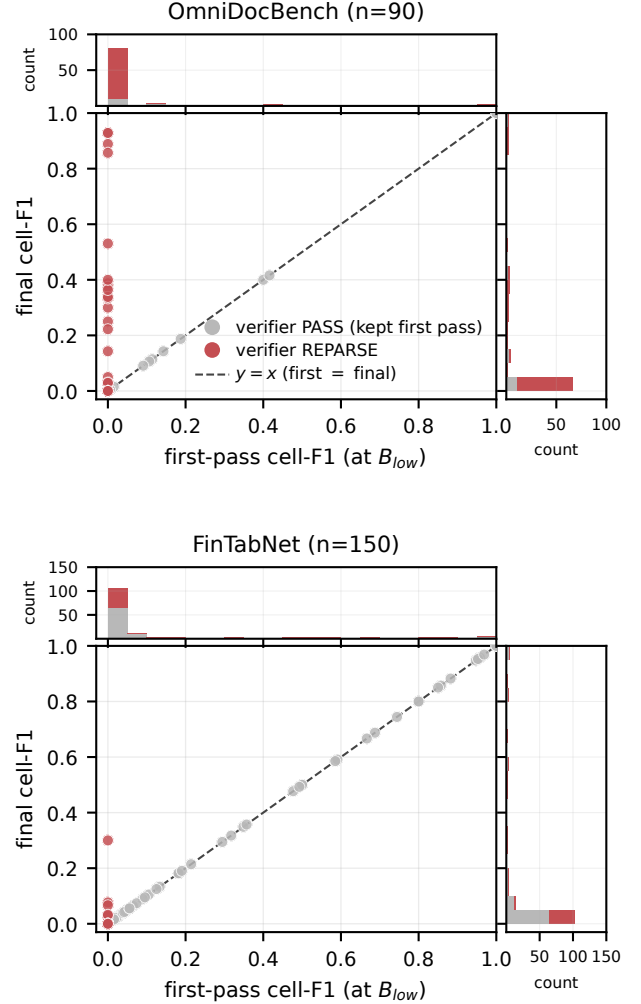


Fig. 3. Per-page first-pass vs. final cell-F1 for the adaptive system, colored by the verifier’s actual decision. Gray (PASS) pages lie on the diagonal by construction; the first pass is kept. Red (REPARSE) pages above the diagonal gained from the second pass; most OmniDocBench gains start from a first-pass score of zero, consistent with the NO_TABLE_FOUND-dominated trigger profile. No page fell below the diagonal in these runs. Stacked marginal histograms (top: first-pass, right: final; same color coding) expose the mass that overplots at the origin: 78/90 OmniDocBench and 93/150 FinTabNet first passes score zero, dropping to 58 and 87 at the final margin as escalation lifts 20 and 6 pages off the floor (cf. Table IV). Points are not jittered, so the diagonal relation is exact.

drives large gains on simple and complex pages, exactly the regime where a selective policy has nothing to offer. The very-complex bucket (51 and 91 pages) sits near zero for every 2B configuration and explains the mass of tied pairs in Table III: on most held-out pages, no 2B policy choice moves the score at all. Absolute extraction quality on these benchmarks is capped by the backbone, not by the routing policy; our claims are about relative allocation at matched cost.

F. Tile budget is not latency

Table VI reports what the tile-based accounting hides. At near-equal tile cost, the adaptive system took $1.7\times$ the fixed-

TABLE V

MACRO CELL-F1 BY DIFFICULTY BUCKET (HELD-OUT), REPORTED AS MEAN [95% BOOTSTRAP CI] (PERCENTILE, 1000 RESAMPLES) OVER THE PER-PAGE SCORES IN EACH BUCKET. THE VERY-COMPLEX BUCKET DOMINATES BOTH SPLITS AND FLOORS NEAR ZERO FOR EVERY 2B CONFIGURATION; EVEN THE 8B REFERENCE REACHES ONLY 0.06–0.07 THERE. TIES ON THESE PAGES ARE WHY THE PAIRED RANK TESTS HAVE FEW INFORMATIVE PAIRS; THE WIDE INTERVALS ON THE SMALL SIMPLE/COMPLEX BUCKETS REFLECT THEIR FEW PAGES.

Dataset	Bucket (n)	low	matched	adaptive	8B
OmniDoc	simple (12)	0.179 [0.000, 0.429]	0.244 [0.065, 0.476]	0.228 [0.050, 0.464]	0.337 [0.118, 0.581]
OmniDoc	complex (27)	0.033 [0.007, 0.068]	0.240 [0.131, 0.360]	0.256 [0.144, 0.374]	0.379 [0.242, 0.520]
OmniDoc	very complex (51)	0.009 [0.000, 0.025]	0.027 [0.004, 0.059]	0.024 [0.004, 0.048]	0.072 [0.028, 0.128]
FinTab	simple (19)	0.190 [0.075, 0.326]	0.142 [0.041, 0.268]	0.194 [0.080, 0.329]	0.202 [0.079, 0.340]
FinTab	complex (40)	0.307 [0.193, 0.423]	0.299 [0.189, 0.411]	0.307 [0.193, 0.424]	0.275 [0.162, 0.397]
FinTab	very complex (91)	0.040 [0.014, 0.070]	0.036 [0.012, 0.067]	0.044 [0.019, 0.076]	0.057 [0.028, 0.091]

TABLE VI

MEASURED WALL-CLOCK SECONDS PER PAGE (MEAN OVER THE HELD-OUT SPLIT, SINGLE L4, BATCH SIZE 1). MATCHED TILE BUDGETS DO NOT PRODUCE MATCHED LATENCY: TWO-PASS SYSTEMS PAY PER-PASS OVERHEADS, AND RUNAWAY DECODES TO THE 2,048-TOKEN CAP INFLATE MEANS.

System	OmniDocBench	FinTabNet
fixed 2B low	27.0	17.6
fixed 2B matched	29.7	17.6
adaptive 2B	52.0	24.9
random 2B (seed range)	49.3–53.7	21.4–25.1
fixed 8B matched	76.2	40.9

matched system’s wall-clock on OmniDocBench (52.0 vs. 29.7 s/page) and $1.4\times$ on FinTabNet (24.9 vs. 17.6), because escalated pages run two full passes and hard pages decode long. Any deployment argument for this policy must therefore be made in tiles (visual-encoder compute) or in batch-throughput settings where latency amortizes, not in single-page latency. We consider publishing both numbers a requirement for this kind of paper rather than a weakness of the method: a matched-cost claim is only as meaningful as the cost unit it is matched on.

VI. DISCUSSION

When the signal helps. The two datasets bracket the answer. FinTabNet’s budget-response curve is flat: uniform tiles beyond 6 buy nothing (Fig. 2, right), so a policy that holds most pages at 6 tiles and escalates the 27% that are structurally broken outperforms every uniform allocation of the same average budget, and outperforms random escalation at the same rate because the broken pages are not random. OmniDocBench inverts every one of those conditions: the curve is steep, the low budget is so low that 78% of first passes fail verification, and escalation degenerates toward uniform spending. The data are consistent with a simple rule of thumb. Verifier-gated escalation is worth considering when uniform budget increases have weak returns and the structural failure rate at low budget is well below one. But two datasets cannot establish that rule as a law, and we present it as the hypothesis our results support, not as a proven mechanism. A controlled experiment that varies budget spread at a fixed escalation rate would test it directly; we did not run one.

The 8B reading. On FinTabNet the adaptive 2B system’s cell-F1 (0.133) is at the level of the 8B model at the same tile budget (0.134). We do not read this as parity. The 8B model is clearly stronger on TEDS (0.527 vs. 0.492), clearly stronger on OmniDocBench (0.200 vs. 0.121), and the tile-matched comparison already concedes it roughly four times the per-token compute. The defensible statement is narrower: on the dataset where routing helps, a 2B model with a structural router closed the cell-F1 gap to its 8B sibling at matched tile budget, while costing about 60% of the 8B wall-clock (24.9 vs. 40.9 s/page).

Publishing the null. Adaptive-inference papers usually report the dataset where the policy wins. The OmniDocBench null is informative in a way the FinTabNet win is not: it identifies the failure mode of verifier gating (escalation saturation) from observables available *before* deployment, namely the structural failure rate at the chosen low budget. A practitioner can measure that rate on unlabeled pages with the same verifier and budget sweep we used, at calibration cost only.

Limitations. The evaluation is 240 held-out pages over two datasets and one backbone family; the confirmatory result is one significant contrast on one dataset. Cell-F1 near zero on very-complex pages means most pages cannot express a policy difference, which both weakens the rank tests (13–51 informative pairs) and caps how much any routing policy could matter here; a stronger backbone might change either side of that. The verifier checks structure only; a fluent but wrong table passes, so the policy inherits the backbone’s semantic errors. The random control shares the adaptive system’s expected cost but not its per-page cost distribution. Budgets were frozen against a 10-tile target, but integer budgets cannot match a fractional realized cost exactly; the over-budget baseline mitigates, and does not eliminate, this mismatch. Finally, the loop-stop truncation, while validated as decision-neutral on calibration data, is an adapter-level intervention a strict replication should reproduce.

VII. CONCLUSION

We evaluated a deterministic structural verifier as a routing signal for adaptive VLM table extraction under a matched-tile-budget protocol with fixed, random, and 8B controls. On FinTabNet, where uniform budget increases have weak returns, verifier-gated escalation outperformed random escalation at

equal expected cost, the one contrast that survives multiple-comparison correction, and held a consistent +0.014 cell-F1 margin over fixed baselines spending either slightly less or slightly more than it. On OmniDocBench, where uniform increases pay and the verifier flags 78% of low-budget pages, the policy added nothing measurable. Structural verification costs milliseconds, requires no training, and leaves an audit trail; these results support using it as an escalation gate where budget-response curves are flat, and measuring the low-budget structural failure rate before assuming it will transfer.

REPRODUCIBILITY

All experiments are driven by versioned configuration files; budgets, prompts, split page-ID manifests, and per-page run logs (including verifier decisions, failure codes, and stage timings) are frozen artifacts, and the analysis stage is deterministic given those inputs. The harness, configurations, frozen artifacts, and per-page logs will be released on publication.

AI USE DISCLOSURE

In accordance with the venue’s policy on generative AI: we used a large language model, under our direction, to assist in drafting and revising manuscript text and analysis scripts. All experimental design, data collection, results, and claims were produced and verified against the project’s experiment artifacts by us, who take full responsibility for the content.

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